

# Improper integrals

## Basic skills

Q1. a)  $\int_0^{\infty} e^{-x} dx$  improper as  $\infty$  limit

b)  $\int_0^{10} \frac{1}{x-4} dx$  improper as undefined at  $x=4$

c)  $\int_0^5 \frac{x-4}{x+1} dx$  Not improper as no  $\infty$  limit / undefined pts

d)  $\int_{-5}^5 \frac{1}{x^2+5} dx$  Not improper as no  $\infty$  limit / undefined pts

e)  $\int_{-\infty}^{\infty} e^{x^2} dx$  improper as  $\infty$  limit

f)  $\int_4^{-4} \frac{1}{(x-4)(x+4)} dx$  improper as undefined at  $x=4$  (and  $x=-4$ )

Q2. • Limit is  $\infty$  so area under infinite length

• undefined so no value (usually as infinite limit) so no value under infinite height

The limit may or may not be finite so not always

Q3. 
$$\int_2^{\infty} \frac{1}{x} dx = \lim_{k \rightarrow \infty} \int_2^k \frac{1}{x} dx$$
$$= \lim_{k \rightarrow \infty} [\ln x]_2^k = \lim_{k \rightarrow \infty} (\ln k - \ln 2) = \infty$$

so improper integral doesn't exist

### Competency skills

Q4. 
$$\int_4^{10} \frac{1}{\sqrt{x-4}} dx = \lim_{k \rightarrow 4} \int_k^{10} \frac{1}{\sqrt{x-4}} dx$$
$$= \lim_{k \rightarrow 4} \int_k^{10} (x-4)^{-\frac{1}{2}} dx = \lim_{k \rightarrow 4} \left[ 2(x-4)^{\frac{1}{2}} \right]_k^{10}$$
$$= \lim_{k \rightarrow 4} (2\sqrt{6} - 2\sqrt{k-4}) = 2\sqrt{6} \text{ so exists}$$

$$\text{Q5. a) } \int_2^{\infty} \frac{1}{x^2} dx = \lim_{k \rightarrow \infty} \int_2^k \frac{1}{x^2} dx$$

$$= \lim_{k \rightarrow \infty} \left[ -x^{-1} \right]_2^k = \lim_{k \rightarrow \infty} \left( -\frac{1}{k} - -\frac{1}{2} \right) = \frac{1}{2}$$

$$\text{b) } \int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{k \rightarrow 0} \int_k^1 x^{-\frac{1}{2}} dx$$

$$= \lim_{k \rightarrow 0} \left[ 2x^{\frac{1}{2}} \right]_k^1 = \lim_{k \rightarrow 0} (2 - 2\sqrt{k}) = 2$$

$$\text{c) } \int_1^{\infty} \frac{3}{\sqrt[3]{x}} dx = \lim_{k \rightarrow \infty} \int_1^k 3x^{-\frac{1}{3}} dx$$

$$= \lim_{k \rightarrow \infty} \left[ \frac{9}{2} x^{\frac{2}{3}} \right]_1^k = \lim_{k \rightarrow \infty} \left( \frac{9}{2} k^{\frac{2}{3}} - \frac{9}{2} \right) = \infty$$

so doesn't exist

$$\text{d) } \int_2^4 \frac{1}{(x-3)^2} dx = \int_2^3 \frac{1}{(x-3)^2} dx + \int_3^4 \frac{1}{(x-3)^2} dx$$

$$= \lim_{k \rightarrow 3} \int_2^k \frac{1}{(x-3)^2} dx + \lim_{k \rightarrow 3} \int_k^4 \frac{1}{(x-3)^2} dx$$

$$= \lim_{k \rightarrow 3} \left[ -(x-3)^{-1} \right]_2^k + \lim_{k \rightarrow 3} \left[ -(x-3)^{-1} \right]_k^4$$

$$= \lim_{k \rightarrow 3} \left( -\frac{1}{k-3} + 1 \right) + \lim_{k \rightarrow 3} \left( -1 + \frac{1}{k-3} \right)$$

As limits are  $\infty/-\infty$  doesn't exist

Q6.  $\int_0^e \ln x \, dx = \lim_{k \rightarrow 0} \int_k^e \ln x \, dx$

oper as limit  $\lim_{k \rightarrow 0} [x \ln x - x]_k^e$

Parts  
 $u = \ln x \quad v = x$   
 $u = \frac{1}{x} \quad v' = 1$   
 $x \ln x - \int 1 \, dx$

$$= \lim_{k \rightarrow 0} (e \ln e - e - (k \ln k - k))$$

given  $x \ln x \rightarrow 0$   
as  $x \rightarrow 0$

$$= e \ln e - e - (0 - 0)$$

$$= e - e = 0 \quad \text{exists}$$

Q7.  $\int_0^\infty \frac{e^{-x}}{1+e^{-x}} \, dx = \lim_{k \rightarrow \infty} \int_0^k \frac{e^{-x}}{1+e^{-x}} \, dx$

$$= \lim_{k \rightarrow \infty} \int_2^{1+\bar{e}^k} \frac{1}{u} \, du$$

$u = 1 + e^{-x}$   
 $du = -e^{-x} \, dx$

$$= \lim_{k \rightarrow \infty} [\ln u]_2^{1+\bar{e}^k} = \lim_{k \rightarrow \infty} (\ln(1+\bar{e}^k) - \ln 2)$$

$$= \ln 1 - \ln 2 = -\ln 2 \quad \text{so exists}$$

Q8.  $\int_1^4 \frac{1}{(x-2)(x-3)} dx$

$$\frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} \Rightarrow \begin{aligned} A+B &= 0 \\ -3A-2B &= 1 \end{aligned}$$

$$= \frac{-1}{x-2} + \frac{1}{x-3} \quad \text{so} \quad \begin{aligned} B &= 1 \\ A &= -1 \end{aligned}$$

$$\text{so } \int_1^4 \frac{1}{(x-2)(x-3)} dx = \int_1^4 \frac{-1}{x-2} dx + \int_1^4 \frac{1}{x-3} dx$$

$$= \int_1^2 \frac{-1}{x-2} dx + \int_2^4 \frac{-1}{x-2} dx$$

$$+ \int_1^3 \frac{1}{x-3} dx + \int_3^4 \frac{1}{x-3} dx$$

$$= \lim_{k \rightarrow 2} [-\ln(x-2)]_1^k + \lim_{k \rightarrow 2} [-\ln(x-2)]_k^4$$

$$+ \lim_{k \rightarrow 3} [\ln(x-3)]_1^k + \lim_{k \rightarrow 3} [\ln(x-3)]_k^4$$

As  $-\ln(k-2) \rightarrow \infty$  as  $k \rightarrow 2$  this

means the entire sum diverges

$$\text{Q9. } \int_0^{\infty} x e^{-x} dx = \lim_{k \rightarrow \infty} \int_0^k x e^{-x} dx$$

$$= \lim_{k \rightarrow \infty} \left[ -x e^{-x} - e^{-x} \right]_0^k$$

Parts  
 $u = x \quad v = -e^{-x}$   
 $u' = 1 \quad v' = e^{-x}$

$$= \lim_{k \rightarrow \infty} \left( -k e^{-k} - e^{-k} - (0 - 1) \right)$$

Now need behaviour of  $k e^{-k}$

$$\lim_{k \rightarrow \infty} \frac{k}{e^k} = \lim_{k \rightarrow \infty} \frac{1}{e^k} = 0$$

L'Hopital's

$$\text{So } \int_0^{\infty} x e^{-x} dx = 0 - 0 + 1 = 1$$

$$\text{Q10. As } \int \frac{1}{\sqrt{x^3}} dx = \int x^{-\frac{3}{2}} dx = -2x^{-\frac{1}{2}}$$

has behaviour of  $\rightarrow 0$  as  $x \rightarrow \infty$

So will exist

$$\text{As } \int \frac{1}{\sqrt[3]{x^2}} dx = \int x^{-\frac{2}{3}} dx = 3x^{\frac{1}{3}}$$

has behaviour of  $\rightarrow \infty$  as  $x \rightarrow \infty$

so will not exist

Q 11. a)  $\int_{-\infty}^{\infty} \frac{1}{x^2} dx$  is improper as  
lower limit  $-\infty$ , upper limit  $\infty$ , undefined at  $x=0$

b) As  $\int \frac{1}{x^2} dx = -\frac{1}{x}$   
has behaviour of  $\rightarrow \infty$  as  $x \rightarrow 0$   
the limit will not exist  
Although  $x \rightarrow \pm \infty$  is fine as behaviour  
is  $\rightarrow 0$

Q 12.  $\int_0^{\infty} \frac{1}{x+s} - \frac{1}{x+1} dx$

$$= \lim_{k \rightarrow \infty} \left[ \ln(x+s) - \ln(x+1) \right]_0^k$$
$$= \lim_{k \rightarrow \infty} \left[ \ln\left(\frac{x+s}{x+1}\right) \right]_0^k = \lim_{k \rightarrow \infty} \left( \ln\left(\frac{k+s}{k+1}\right) - \ln s \right)$$
$$= \lim_{k \rightarrow \infty} \left( \ln\left(\frac{1+\frac{s}{k}}{1+\frac{1}{k}}\right) - \ln s \right) = \ln 1 - \ln s$$
$$= -\ln s$$

Q13.  $\int_{-\infty}^{\infty} x \, dx = \int_0^{\infty} x \, dx + \int_{-\infty}^0 x \, dx$

As  $\int x \, dx = \frac{1}{2} x^2 \, dx$  and  $x^2 \rightarrow \infty$  as  $x \rightarrow \infty$

so integral does not exist

but the cauchy value Principal does agree as it has

$$\begin{aligned} \int_{-\infty}^{\infty} x \, dx &= \lim_{a \rightarrow \infty} \int_a^{-a} x \, dx \\ &= \lim_{a \rightarrow \infty} \left[ \frac{1}{2} x^2 \right]_a^{-a} = \lim_{a \rightarrow \infty} 0 = 0 \end{aligned}$$

so exists as cauchy principle value